ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET Mechatronic Systems paper (project based, 100 marks): design brief and specifications, automating a jar-capping station, gear-train calculations, modifying a control circuit, valve symbols, budgets, Gantt charts, risk assessment, system block diagrams, testing and commissioning, fault-finding, maintenance strategies and MTBF.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers.
5. Draw sketches, charts and circuits neatly. Use only a black or blue pen.

> **PROJECT.** A small food company closes glass jars of jam by screwing on the caps by hand. The owner wants a capping station added to the existing conveyor: a sensor detects each jar, a pneumatic stop gate holds the jar under the capping head, a cylinder lowers the head, and a DC gear motor turns the head to screw the cap on. The stop gate then opens and the jar moves on. A counter keeps track of the number of jars capped.

QUESTION 1

1.1 Explain the difference between a design brief and a design specification. (4)

1.2 Write SIX specifications for the capping station. (6)

[10]

QUESTION 2

2.1 Make a neat freehand sketch of the capping station. Show the conveyor, the jar sensor, the stop gate, the lowering cylinder and the capping head with its motor. (5)

2.2 FIGURE 1 shows the gear train between the motor and the capping head. The motor runs at 1 400 r/min and delivers 0,18 kW.

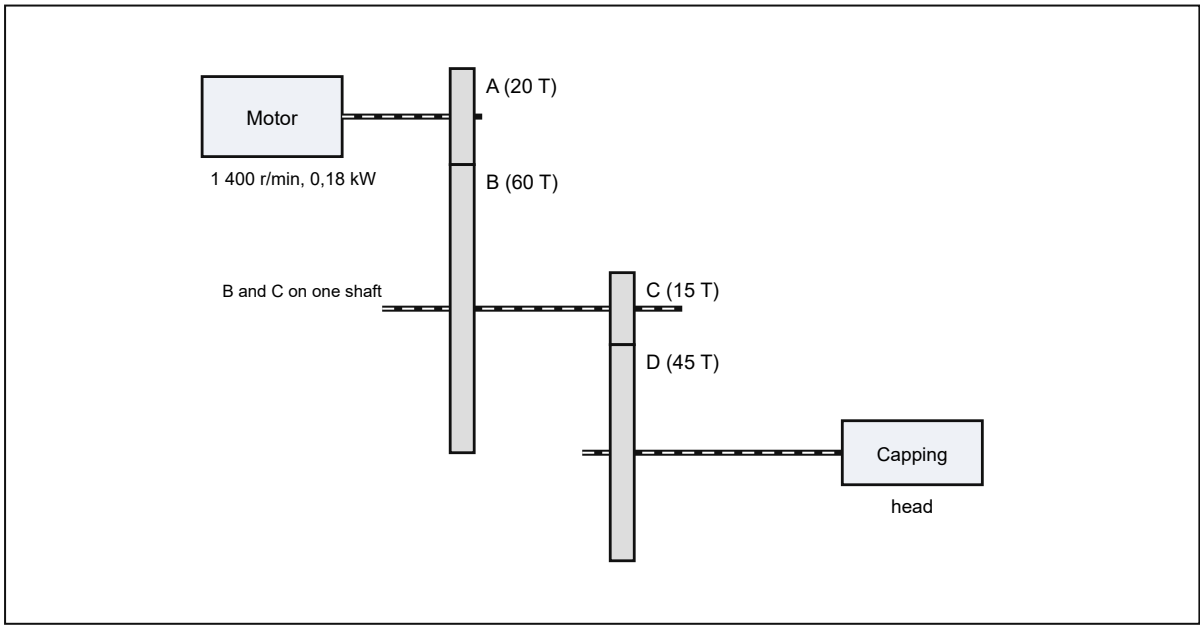


FIGURE 1: Gear train of the capping head

2.2.1 Calculate the total gear ratio. (2)

2.2.2 Calculate the speed of the capping head. (2)

2.2.3 Calculate the torque at the capping head if the gear train is 85% efficient. (4)

2.3 The conveyor is switched on and off by the circuit in FIGURE 2. Redraw and modify it so that, while the conveyor is switched on, jar sensor S1 energises the stop gate solenoid Y1, and the lowering solenoid Y2 and capping motor M2 then operate for 4 seconds (timer T1). (8)

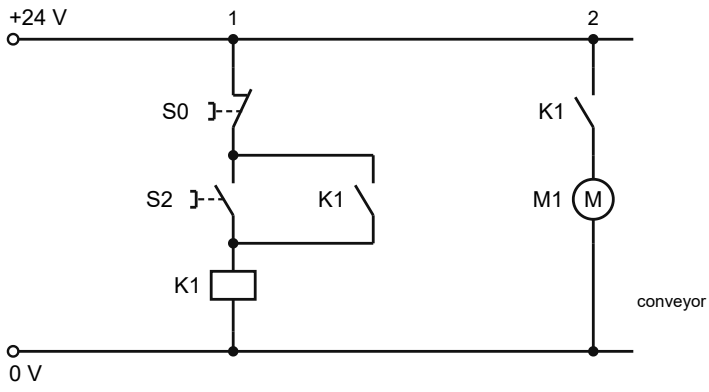


FIGURE 2: Existing conveyor circuit

2.4 Draw the symbol of a 5/2-way valve operated by a solenoid, with a spring return. (2)

2.5 Give TWO reasons why a slipping (torque-limiting) clutch is fitted between the motor and the capping head. (2)

[25]

QUESTION 3

3.1 Draw up a budget for the capping station with suitable column headings and at least SIX items. (8)

3.2 Draw a Gantt chart for SIX activities of the project over a period of 4 weeks. Some activities must overlap. (10)

3.3 Name THREE hazards during the installation of the capping station and give ONE control measure for EACH. (6)

3.4 Draw a system block diagram of the capping station, showing its inputs, controller and outputs. (6)

[30]

QUESTION 4

4.1 Explain the difference between testing and commissioning. (2)

4.2 Use a spider diagram to show SIX areas that must be tested on the capping station. (6)

4.3 Explain how the tightness of the caps is checked and adjusted during commissioning. (3)

4.4 List FOUR documents that must be handed to the owner when the station is handed over. (4)

[15]

QUESTION 5

5.1 The capping head does not move down, although the PLC output LED for Y2 is on. Describe a logical fault-finding procedure. (6)

5.2 Compare breakdown, preventive and predictive maintenance by giving ONE advantage of EACH. (6)

5.3 The station ran for 720 hours in a month. It broke down 4 times and the repairs took 12 hours altogether. Calculate the:

5.3.1 mean time between failures (MTBF) (2)

5.3.2 mean time to repair (MTTR) (2)

5.3.3 availability of the station as a percentage (2)

5.4 Explain why accurate maintenance records must be kept. (2)

[20]

TOTAL: 100

END OF QUESTION PAPER